

PRAKTIKUM BASIS DATA

MODUL 13
NoSQL_Document_Database



**Universitas
Telkom**

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

1.



```
MongoDB Compass - localhost:27017/Shell
Connections Edit View Help

Compass
My Queries
Data Modeling

CONNECTIONS (1)
Search connections
localhost:27017
  admin
  config
  local
  toko_0802
    toko

Welcome toko mongosh localhost:27017

> use toko
switched to db toko

> db.products.find()
{
  "_id": "ObjectID('6a26a3b6f3b71e437ff6a5c8')",
  "name": "Toko Berkah",
  "category": "Elektronik",
  "city": "Jakarta",
  "address": {
    "street": "Jl. Sudirman",
    "zipcode": "10220"
  },
  "products": [
    {
      "item": "TV",
      "price": 3500000,
      "stock": 10
    },
    {
      "item": "Kulkas",
      "price": 2800000,
      "stock": 5
    }
  ]
}
```

4.

```
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CONNECTIONS (1)
Search connections
localhost:27017
  admin
  config
  local
  toko_0802
    toko

Welcome toko mongosh localhost:27017

> db.toko.updateOne(
  { name: "Toko Berkah" },
  { $set: { category: "Aksesoris" } })
{
  "acknowledged": true,
  "insertedId": null,
  "matchedCount": 1,
  "modifiedCount": 1,
  "upsertedCount": 0
}

> db.toko.updateOne(
  { name: "Toko Maju" },
  {
    $push: {
      products: {
        item: "House Gaming",
        price: 450000,
        stock: 8
      }
    }
  })
{
  "acknowledged": true,
  "insertedId": null,
  "matchedCount": 0,
  "modifiedCount": 0,
  "upsertedCount": 0
}

> db.toko.find()
{
  "_id": "ObjectID('6a26a3b6f3b71e437ff6a5c8')",
  "name": "Toko Alfa Jaya",
  "category": "Fashion",
  "city": "Purwokerto",
  "address": {
    "street": "Jl. Sudirman",
    "zipcode": "53111"
  },
  "products": [
    {
      "item": "Noodle",
      "price": 150000,
      "stock": 12
    },
    {
      "item": "Sepatu",
      "price": 300000,
      "stock": 7
    }
  ]
}
```

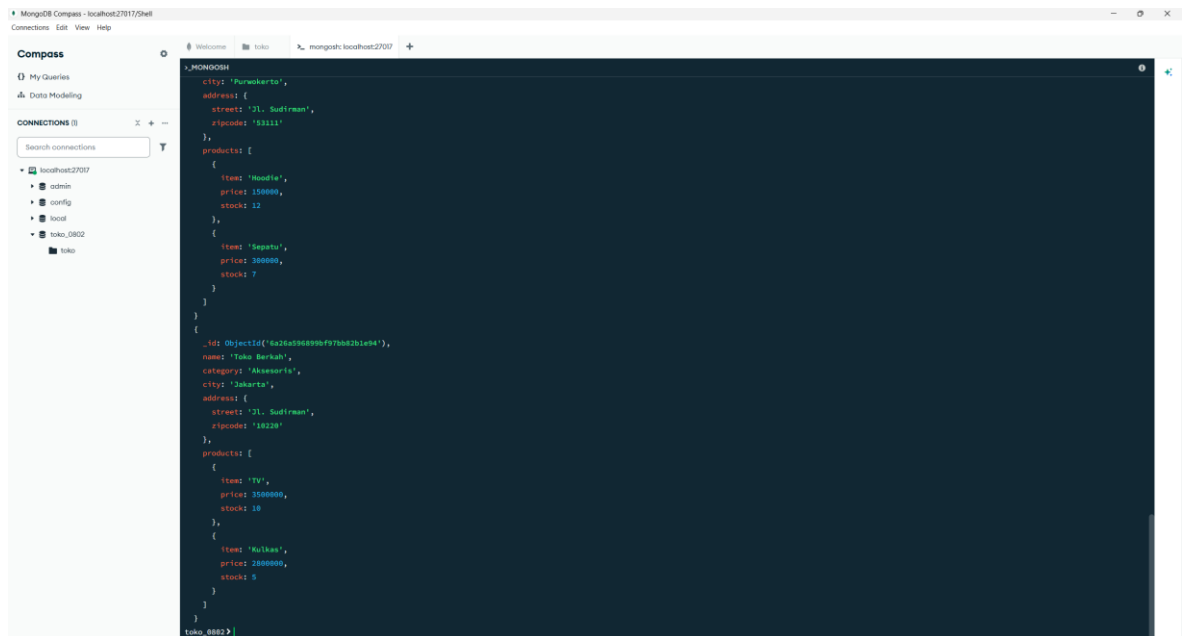
```
MongoDB Compass - localhost:27017/Shell
Connections Edit View Help

Compass
My Queries
Data Modeling

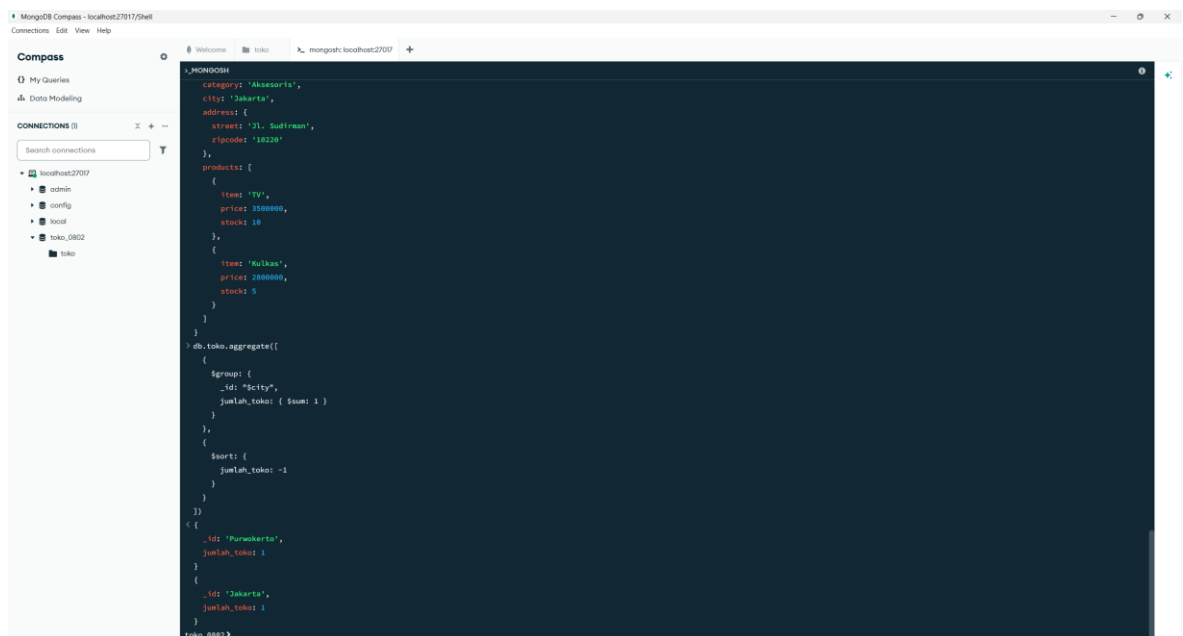
CONNECTIONS (1)
Search connections
localhost:27017
  admin
  config
  local
  toko_0802
    toko

Welcome toko mongosh localhost:27017

> db.toko.find()
{
  "_id": "ObjectID('6a26a3b6f3b71e437ff6a5c8')",
  "name": "Toko Alfa Jaya",
  "category": "Fashion",
  "city": "Purwokerto",
  "address": {
    "street": "Jl. Sudirman",
    "zipcode": "53111"
  },
  "products": [
    {
      "item": "Noodle",
      "price": 150000,
      "stock": 12
    },
    {
      "item": "Sepatu",
      "price": 300000,
      "stock": 7
    }
  ]
},
{
  "_id": "ObjectID('6a26a3b6f3b71e437ff6a5c8')",
  "name": "Toko Berkah",
  "category": "Aksesoris",
  "city": "Jakarta",
  "address": {
    "street": "Jl. Sudirman",
    "zipcode": "10220"
  },
  "products": [
    {
      "item": "TV",
      "price": 3500000,
      "stock": 10
    },
    {
      "item": "Kulkas",
      "price": 2800000,
      "stock": 5
    }
  ]
}
```



5.



6.

The screenshot shows the MongoDB Compass interface with the mongo shell open. The left sidebar shows the 'CONNECTIONS' tab with a search bar and a list of connections: 'localhost:27027' (expanded) showing 'admin', 'config', 'local', and 'toko_0802', and 'toko'. The main window displays the following aggregation pipeline in the mongo shell:

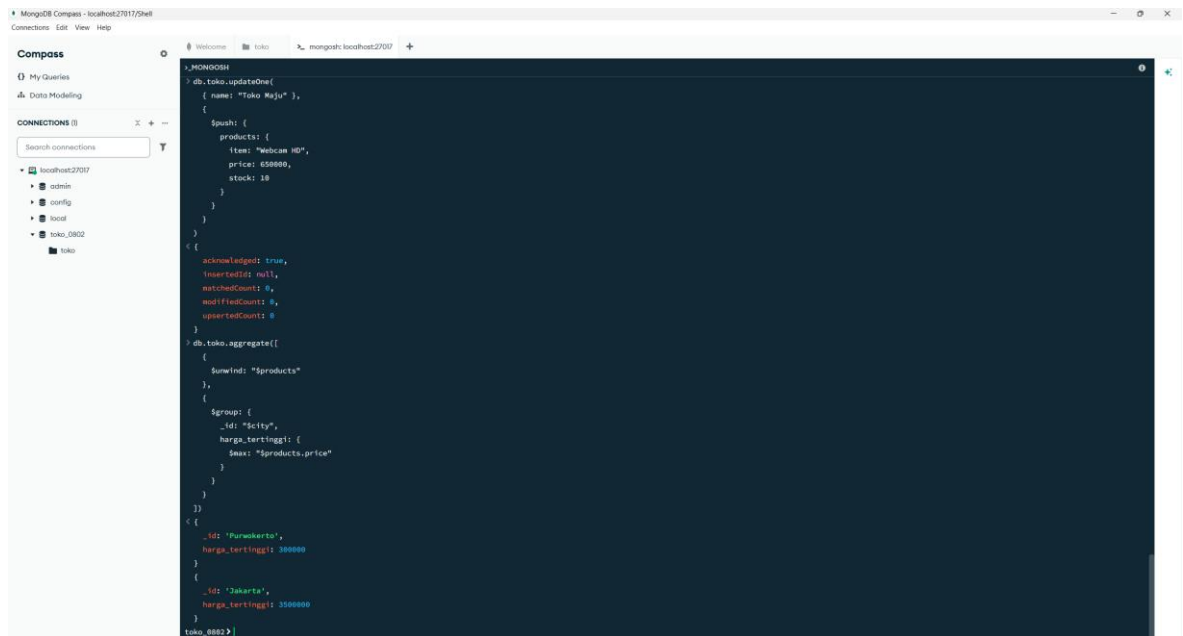
```
> use toki
> db.toko.aggregate([
  {
    $sort: {
      jumlah_toko: -1
    }
  },
  {
    $group: {
      _id: "Purwokerto",
      jumlah_toko: 1
    }
  },
  {
    $group: {
      _id: "Jakarta",
      jumlah_toko: 1
    }
  },
  {
    $sort: {
      rata_rata_harga: -1
    }
  },
  {
    $group: {
      _id: "Aksesoris",
      rata_rata_harga: { $avg: "$products.price" }
    }
  },
  {
    $group: {
      _id: "Fashion",
      rata_rata_harga: { $avg: "$products.price" }
    }
  },
  {
    $sort: {
      rata_rata_harga: -1
    }
  },
  {
    $group: {
      _id: "Aksesoris",
      rata_rata_harga: { $avg: "$products.price" }
    }
  },
  {
    $group: {
      _id: "Fashion",
      rata_rata_harga: { $avg: "$products.price" }
    }
  }
])
```

7.

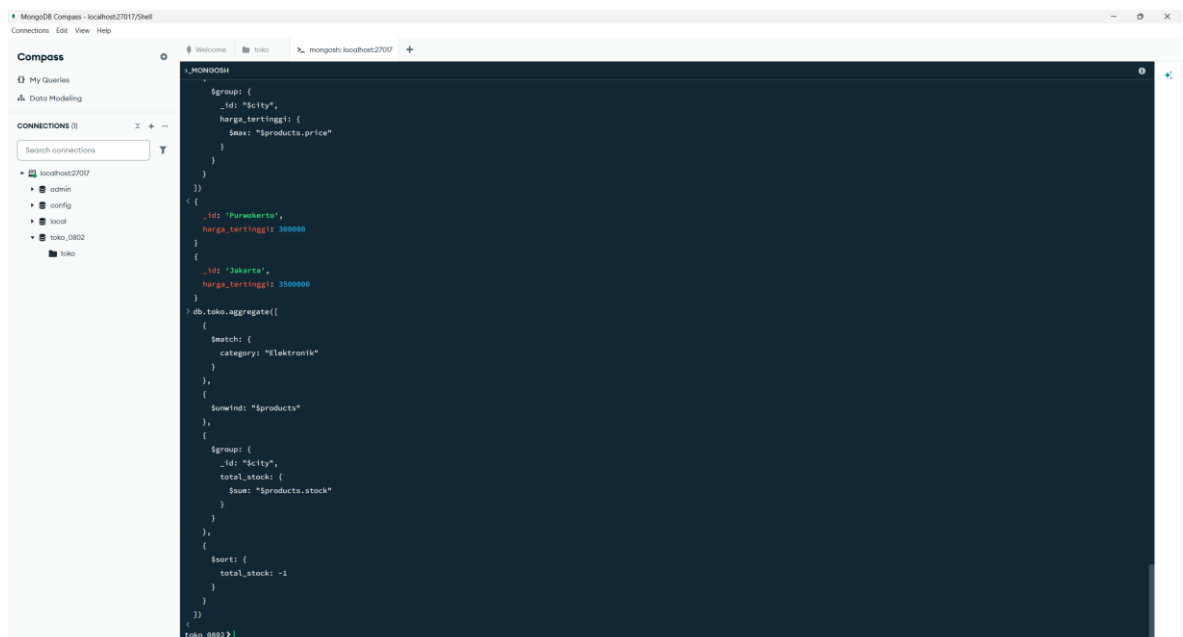
The screenshot shows the MongoDB Compass interface with the mongo shell open. The left sidebar shows the 'CONNECTIONS' tab with a search bar and a list of connections: 'localhost:27027' (expanded) showing 'admin', 'config', 'local', and 'toko_0802', and 'toko'. The main window displays the following updateOne operation in the mongo shell:

```
> use toki
> db.toko.updateOne(
  { name: "Toko Maju" },
  {
    $push: {
      products: {
        item: "Medicine HPD",
        price: 650000,
        stock: 10
      }
    }
  },
  {
    acknowledge: true,
    insertedId: null,
    matchedCount: 0,
    modifiedCount: 0,
    upsertedCount: 0
  }
)
```

8.



9.



10.

